

Basic Concepts
Operations and Laws
Products, Relations, Functions
Partitions and Disjointification
Covers and Orderings

Sets

Ali Karbala

Lebanese University

Reference: Theory of Statistics, James E. Gentle

Outline I

- 1 Basic Concepts
- 2 Operations and Laws
- 3 Products, Relations, Functions
- 4 Partitions and Disjointification
 - Theorem 0.0.1
 - Corollary 0.0.1.1
 - Intuition
- 5 Covers and Orderings
 - Partial and Linear Orderings
 - Well-Ordered Sets

Outline

- 1 Basic Concepts
- 2 Operations and Laws
- 3 Products, Relations, Functions
- 4 Partitions and Disjointification
 - Theorem 0.0.1
 - Corollary 0.0.1.1
 - Intuition
- 5 Covers and Orderings
 - Partial and Linear Orderings
 - Well-Ordered Sets

Basic Concepts

- 1 A set is a collection of elements.
- 2 $A_2 \subseteq A_1$: A_2 is a **subset** of A_1 .
- 3 $A_2 \subset A_1$: A_2 is a **proper subset** of A_1 .
- 4 $A_1 \cup A_2$: **union**, all elements in A_1 or A_2 .
- 5 $A_1 \cap A_2$: **intersection**, elements in both sets.
- 6 Union and intersection are commutative: $A_1 \cup A_2 = A_2 \cup A_1$,
 $A_1 \cap A_2 = A_2 \cap A_1$.
- 7 $\emptyset$: the **empty set**, contains no elements.

Cardinality

- ⑧ The **cardinality** of a set is the number of its elements.
- ⑨ Finite sets: cardinality is an integer.
- ⑩ A set is **countable** if it can be put in one-to-one (injective) correspondence with $\mathbb{N}$.
- ⑪ A set that is countable but infinite is **countably infinite**.
- ⑫ Any interval of $\mathbb{R}$ is **uncountably infinite** (continuum).

Sample Space and Complements

- 13 **Sample Space:** the universal set of all outcomes.
- 14 If A is the universal set, then $A_1 \subseteq A$.
- 15 **Complement:** $A_1^c = A - A_1$.
- 16 **Difference:** $A_1 - A_2 = A_1 \cap A_2^c$.
- 17 **Proper difference:** $A_1 - A_2$ if $A_2 \subseteq A_1$.
- 18 **Symmetric difference:** $A_1 \triangle A_2 = (A_1 - A_2) \cup (A_2 - A_1)$
(0.0.1).

Outline

- 1 Basic Concepts
- 2 **Operations and Laws**
- 3 Products, Relations, Functions
- 4 Partitions and Disjointification
 - Theorem 0.0.1
 - Corollary 0.0.1.1
 - Intuition
- 5 Covers and Orderings
 - Partial and Linear Orderings
 - Well-Ordered Sets

De Morgan's Laws

- 19 Symmetric difference is commutative: $A_1 \triangle A_2 = A_2 \triangle A_1$.
- 20 **De Morgan's Laws:** $(A_1 \cup A_2)^c = A_1^c \cap A_2^c$ (0.0.2),
 $(A_1 \cap A_2)^c = A_1^c \cup A_2^c$ (0.0.3).
- 21 Extended De Morgan's Laws (if A_i is countable $\forall i$ in $\mathbb{N}$):

$$\left(\bigcup_{i=1}^{\infty} A_i \right)^c = \bigcap_{i=1}^{\infty} A_i^c, \quad \left(\bigcap_{i=1}^{\infty} A_i \right)^c = \bigcup_{i=1}^{\infty} A_i^c$$

- 22 The **smallest subset** of a universal set A is the intersection of all subsets of A with a given property.

Outline

- 1 Basic Concepts
- 2 Operations and Laws
- 3 **Products, Relations, Functions**
- 4 Partitions and Disjointification
 - Theorem 0.0.1
 - Corollary 0.0.1.1
 - Intuition
- 5 Covers and Orderings
 - Partial and Linear Orderings
 - Well-Ordered Sets

Product Sets

- 23 $A \times B$, The **cartesian product** (or direct product or cross product) of two sets A and B , is the set of all doubletons, (a_i, b_j) , where $a_i \in A$ and $b_j \in B$. The set $A \times B$ is called a product set.
- 24 $A \times B \neq B \times A$ unless $B = A$.
- 25 $\emptyset \times A = A \times \emptyset = \emptyset = \emptyset \times \emptyset$.

Relations and Functions

- 26 A **relation** is a set of doubletons, or pairs of elements; that is, a relation is a subset of a cartesian product of two sets.
- 27 A **function** (transform or an operator) is a relation in which no two different pairs have the same first element.

Functions (continued)

- 28 To say that f is a function from Ω (**domain**) to Λ (**range**), written $f : \Omega \rightarrow \Lambda$, means that for every $\omega \in \Omega$ there is a pair in $f(\omega, f(\omega))$ whose first member is ω (**argument**).
- 29 A function is **onto or surjective** (Surjective) if for every $\lambda \in \Lambda$ there is a pair in f whose second member is λ .

Functions (continued)

- 30 functions that are **one-to-one or injective** ($1 : 1$), meaning that no two pairs have the same second member.
- 31 A function that is one-to-one and onto is called a **bijection**.
- 32 A function f that is one-to-one has an inverse, written f^{-1} , that is a function from Λ to Ω , such that if $f(\omega_0) = \lambda_0$, then $f^{-1}(\lambda_0) = \omega_0$.

Outline

- 1 Basic Concepts
- 2 Operations and Laws
- 3 Products, Relations, Functions
- 4 **Partitions and Disjointification**
 - Theorem 0.0.1
 - Corollary 0.0.1.1
 - Intuition
- 5 Covers and Orderings
 - Partial and Linear Orderings
 - Well-Ordered Sets

Partitions; Disjoint Sets

- 32 A **partition** of a set S is a collection of disjoint subsets of S whose union is S .

Theorem 0.0.1

Theorem. Let $A_1, A_2, \dots$ be a sequence of subsets of a universal set S . Define

$$D_1 = A_1, \quad D_n = A_n \setminus \bigcup_{i=1}^{n-1} A_i, \quad n = 2, 3, \dots$$

(0.06) Then:

- The sets D_n are pairwise disjoint.

- $\bigcup_{i=1}^{\infty} D_i = \bigcup_{i=1}^{\infty} A_i.$

Moreover, if

$$D_0 = S \setminus \bigcup_{i=1}^{\infty} A_i,$$

Example

Let $S = \{1, 2, 3, 4, 5\}$ and

$$A_1 = \{1, 2\}, \quad A_2 = \{2, 3\}, \quad A_3 = \{1, 4, 5\}.$$

Then:

$$D_1 = \{1, 2\}, \quad D_2 = \{3\}, \quad D_3 = \{4, 5\}.$$

$$\text{Disjoint: } D_1 \cap D_2 \cap D_3 = \emptyset \qquad \bigcup_{i=1}^3 D_i = \{1, 2, 3, 4, 5\}.$$

Intuition

- Each D_n collects the “new” elements of A_n that were not present before.
 - This construction removes overlap: if $n < m$, then D_m explicitly avoids A_n .
 - The union is preserved because every element of some A_i appears first in a unique D_n .
- ⇒ reorganized into disjoint blocks: D_1, D_2, D_3 .

Importance in Statistics

- Disjointness is crucial in probability theory:

$$P\left(\bigcup_i D_i\right) = \sum_i P(D_i).$$

- Allows us to “partition” complex events into non-overlapping parts.
- Useful in measure theory and integration, since additivity holds only for disjoint sets.
- Forms the foundation for:
 - Probability partitions,
 - Disjoint unions in expectation and variance calculations,

Corollary 0.0.1.1

Corollary 0.0.1.1. Let $\{A_n\}$ be a sequence of sets such that

$$A_1 \subseteq A_2 \subseteq \cdots .$$

Define the sequence $\{D_n\}$ by

$$D_n = A_{n+1} \setminus A_n, \quad n = 1, 2, \dots \quad (0.0.7)$$

Then the sets $\{D_n\}$ are disjoint, and

$$\bigcup_{i=1}^m D_i = \bigcup_{i=1}^m A_i.$$

Disjointification and Inclusion–Exclusion

A useful partition of the union $S = A_1 \cup A_2$ uses three disjoint sets:

$$\{ A_1 \setminus (A_1 \cap A_2), A_2 \setminus (A_1 \cap A_2), (A_1 \cap A_2) \}. \quad (0.0.8)$$

Definition. This process is called **disjointification**, and expression (0.0.8) is the **inclusion–exclusion formula**.

Example: Increasing Sets and Disjointification

Example.

Let

$$A_1 = \{1, 2\}, \quad A_2 = \{1, 2, 3, 4\}, \quad A_3 = \{1, 2, 3, 4, 5, 6\}.$$

Then the disjoint pieces are

$$D_1 = A_1 = \{1, 2\}, \quad D_2 = A_2 \setminus A_1 = \{3, 4\}, \quad D_3 = A_3 \setminus A_2 = \{5, 6\}.$$

Thus,

$$A_1 = D_1, \quad A_2 = D_1 \cup D_2, \quad A_3 = D_1 \cup D_2 \cup D_3,$$

and in general

$$\bigcup_{i=1}^n A_i = \bigcup_{i=1}^n D_i.$$

Intuition

Intuition.

- Think of A_n as a collection growing over time.
- Each D_n represents the *new elements* added at stage n .
- No two stages add the same element twice, hence disjoint.
- The full union is just the pile of all these “new arrivals.”

Uses in Statistics

Importance in Statistics.

- **Measure theory:** Basis for continuity from below:

$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) = \lim_{n \rightarrow \infty} \mu(A_n).$$

- **Probability:** For increasing events $A_n \uparrow A$,

$$P(A) = \lim_{n \rightarrow \infty} P(A_n).$$

- **Applications:** Borel–Cantelli lemmas, stopping times, and the inclusion–exclusion principle in probability.

Outline

- 1 Basic Concepts
- 2 Operations and Laws
- 3 Products, Relations, Functions
- 4 Partitions and Disjointification
 - Theorem 0.0.1
 - Corollary 0.0.1.1
 - Intuition
- 5 **Covers and Orderings**
 - Partial and Linear Orderings
 - Well-Ordered Sets

Covering Sets

- 33 A collection of sets $\mathcal{A}$ is said to **cover** a set S if

$$S \subseteq \bigcup_{A_i \in \mathcal{A}} A_i.$$

- 34 Given a collection $\mathcal{A} = \{A_1, A_2, \dots\}$ that covers a set S , a **partition** of S can be formed by removing some of the intersections of the sets in $\mathcal{A}$.

Partial and Linear Orderings

35 A set A is said to be **partially ordered** if there exists a relation $\leq$ on $A \times A$ such that:

- $\forall a \in A, a \leq a$ (reflexive)
- For $a, b, c \in A, a \leq b, b \leq c \Rightarrow a \leq c$ (transitive)
- For $a, b \in A, a \leq b, b \leq a \Rightarrow a = b$ (antisymmetric)

The relation $\leq$ is called an **ordering**. A set together with a partial ordering is called a **poset**.

36 A set A is called **linearly ordered** (or **totally ordered**) if it is partially ordered. and every pair of elements $a, b \in A$ can be compared by the ordering relation.

Ordering in $\mathbb{R}$ and $\mathbb{R}^2$

- The real numbers $\mathbb{R}$ are **linearly ordered** using the common inequality relation $<$.
- This relation can be extended elementwise to define a **partial ordering** in a set $S \subseteq \mathbb{R}^2$.

For example: we cannot use the relation for $(1,2)$ and $(2,1)$, instead, we $1 \leq 2$ and $2 \leq 1$.

Well-Ordered Sets

- 38 A set A is **well-ordered** if it is an ordered set in which every nonempty subset has a *smallest* element.
- 39 (Axiom of Choice) **Every set** (even $\mathbb{R}^d$) admits some well-ordering.
- 40 With the usual $\leq$:
 - $\mathbb{N}$ is **well-ordered**.
 - $\mathbb{Z}$ and $\mathbb{R}$ are **not** well-ordered.

Examples (why not well-ordered under the usual $\leq$).

- $\mathbb{Z}$: the subset $\{\dots, -3, -2, -1\}$ has *no smallest element* (for any n there is $n - 1$).
- $\mathbb{R}$: the subset $(0, 1)$ has *no smallest element* (for any $x \in (0, 1)$, $x/2 \in (0, 1)$ and $x/2 < x$).